

Using the X-ray computed tomography method to predict the saturated hydraulic conductivity of the upper root zone in the Loess Plateau in China

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The recovery of natural vegetation significantly improves soil macroporosity through root decay and biological activity. Macropores are preferential pathways for the movement of water to deep soil. However, the quantification of soil macropore structure and its relationship with soil hydraulic conductivity (K_{sat}) are not well understood. We characterized the macropores under different vegetation types to evaluate the effects of macropores on K_{sat} . Undisturbed soil cores were collected from four treatments: areas dominated by *Quercus liaotungensis* Koidz. (QLI), *Pinus tabulaeformis* Carrière (PTA), *Caragana korshinskii* Kom. (CKO), and *Medicago sativa* L. (MSA). A medical computed tomography (CT) scanner was used to acquire images with a voxel size of 0.977 by 0.977 by 1.000 mm for depths of 0 to 360 mm. We used ImageJ software to quantify the macropore properties. Soil structure and K_{sat} improved with the succession of vegetation. The mean macropore volume fraction across the soil cores for the QLI, PTA, CKO, and MSA treatments were 6.6, 3.5, 1.3, and 0.6% within a 4900-mm² area, respectively. Macropore quantity (volume fraction and number) had a higher R^2 for predicting K_{sat} than macropore morphology (branch density, connectivity density, and junction density). Moreover, the grayscale values were negatively correlated with K_{sat} and accounted for 78.8% of the variation in K_{sat} . Grayscale values, volume fraction, and the number of macropores were the best combination of CT-measured parameters for predicting K_{sat} , accounting for 81.9% of the variation in K_{sat} . The CT-measured parameters could be used to estimate K_{sat} in the upper root zone in the Loess Plateau.

Abbreviations: 3D, three-dimensional; BD, bulk density; CT, computed tomography; CKO, *Caragana korshinskii*; K_{sat} , saturated hydraulic conductivity; MSA, *Medicago sativa*; PTA, *Pinus tabulaeformis*; PVC, Polyvinyl chloride; QLI, *Quercus liaotungensis* K.; SOC, soil organic C.

The Chinese government initiated the Grain for Green Project in 1999 to control severe soil erosion in the Loess Plateau. The mean vegetation coverage across the Loess Plateau reached 60.2% by the end of 2013 (Gao et al., 2017). The recovery of the natural vegetation in particular improves soil macroporosity (Li and Shao, 2006; Li et al., 2016). Macropores (diameter > 1000 μm) (Luxmoore, 1981) are preferential pathways for the movement of most water to deep soil after heavy rains and can help to reduce surface runoff (Pachepsky et al., 2000; Devitt and Smith, 2002; Léonard et al., 2004; Li et al., 2009). Reductions in soil porosity and pore continuity usually imply a negative impact on soil hydraulic properties (Berger and Hager, 2000). The importance of macropores as preferential pathways for soil water has been widely recognized (Mooney and Morris, 2008; Rab et al., 2014). Characterizing the properties of soil macropores is important for maximally exploiting the low rainfall in the Loess Plateau.

Core Ideas

- Computed tomography (CT)-measured properties were in the order: soil under *Q. liaotungensis* > *P. tabulaeformis* > *C. korshinskii* > *M. sativa*.
- The parameters measured by CT could help estimate saturated hydraulic conductivity (K_{sat}) in the Loess Plateau.
- Macropore quantity has a higher R^2 for predicting K_{sat} than the morphology of macropores.

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MATERIALS AND METHODS

Experimental Site and Soil Sample Preparation

We selected the Ziwoiling Forest area in Fu County and the Liudaogou catchment in Shenmu County as sites for sampling soil cores. The Ziwoiling Forest area (33°50'N, 107°30'E; 920–1683 m a.s.l.; Fig. 1) occupies approximately 23,000 km² and is the sole secondary forest region remaining in the Loess Plateau. The climate is semiarid with a mean annual rainfall of 577 mm, 60% of which falls from June to August, and a mean annual air temperature of 9.9°C. The loess layer is ~50 m thick and the soil type is a calcareous loamy soil classified as an entisol. The total forest coverage is over 70%. The climax community stage is dominated by *Q. liaotungensis*. The Liudaogou catchment is located in the northern Loess Plateau and is ~14 km west of Shenmu County, Shaanxi Province, China. This catchment is located at 110°21' to 110°23'E and 38°46' to 38°51'N at an elevation of 1094 to 1274 m a.s.l. The region is characterized by a temperate, semiarid zone with a mean annual precipitation of 430 mm, 77% of which occurs between July and September. The average annual temperature is 8.4°C and the mean annual potential evapotranspiration is 785 mm (Jia et al., 2013). The study area is situated in the center of a water–wind erosion crisscross region that sustains serious soil erosion.

In Ziwoiling, we collected the samples in soils covered by 45-yr-old *Q. liaotungensis* (QLI) and 40-yr-old *P. tabuliformis* (PTA). The density of the forest canopy was >70%, with *Viburnum schensianum* Maxim. dominating the shrub layer, and *Artemisia capillaris* Miq. and *Potentilla chinensis* Ser. dominating the herbaceous layer. A litter layer 2 cm deep covered the soil surface. We collected the Liudaogou catchment samples from soils beneath two representative species [40-yr-old *C. korshinskii* (CKO) and 34-yr-old *M. sativa* (MSA)]. These two dryland perennial species have been established in this region for the last few decades to remedy the severe soil erosion. They are commonly planted on degraded slopes with no fertilization or irrigation, and *M. sativa* is cut once or twice a year to provide livestock fodder. Needlegrass (*Stipa bungeana* Trin.) also occurs sporadically

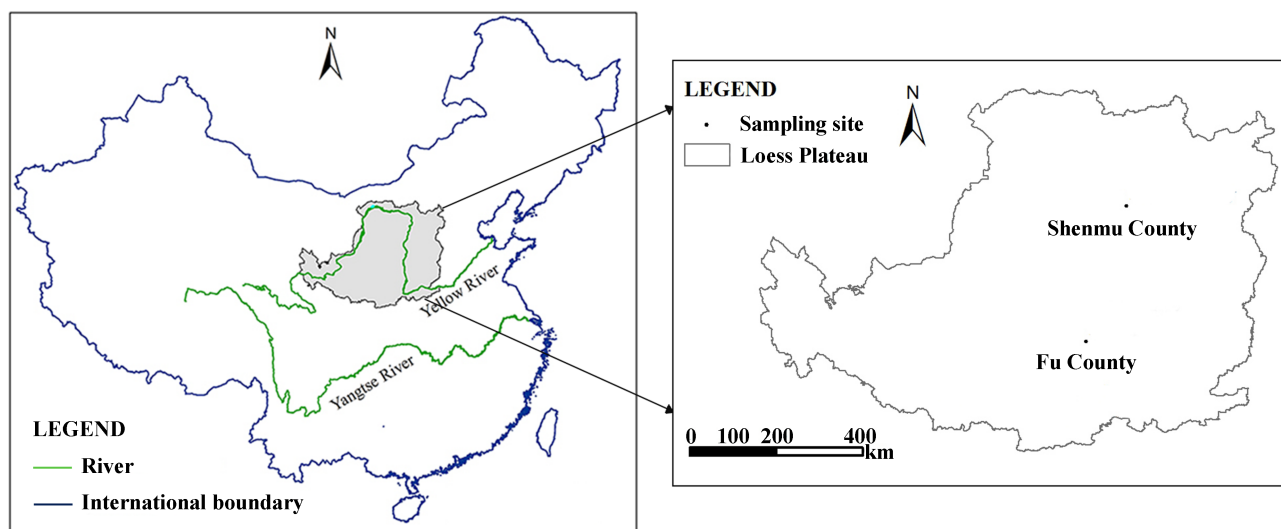


Fig. 1. A map depicting the location of the study area in China and that of the sampling sites in the Loess Plateau.

in areas not covered by the two perennial species. Details of the sample locations and the macropore characteristics of CKO and MSA can be found in the study of Li et al. (2016). During sample collection, the soil surface in the Liudagou catchment had less litter than that in Ziwojing. The study design thus involved four treatments: PTA, QLI, CKO, and MSA.

Undisturbed soil cores were collected 50 cm from a representative plant for each treatment from 27 May to 5 June 2014 after a field survey. Polyvinyl chloride (PVC) cylinders 360 mm long and 110 mm wide with walls 3 mm thick were used to collect the soil samples. Two cores were collected for each treatment because of the high labor cost of core sampling and the processing of scanned images. The samples were collected 50 cm away from the representative plants because the characteristics of soil macropores can vary greatly with distance from plants because of the effects of root growth. The PVC cylinders (with a sharp edge) were hammered vertically and fully into the ground with a plastic hammer and a rubber dumbbell. They were then dug out with a shovel. Two PVC caps and masking tape were used to secure the soil inside the cylinders and protect it during transport. Details of the sample collection were described by Li et al. (2016).

Scanning and Image Analysis

We used a MX16 CT scanner (Philips, Amsterdam, Netherlands) at Yangling Hospital to acquire the CT images. The scan system was set to 140 kV, 200 mA, and 21.4 s scan time. The pixel resolution of the original images was 1.024 pixels per mm. The X-ray beam width or slice thickness was 1 mm. Images with a voxel size of 0.977 by 0.977 by 1.000 mm were produced. The smallest value of pores we obtained was 0.954 mm² (i.e., an equivalent diameter equal to 1.102 mm). The data were then stored for subsequent image analysis. The characteristics of the pores in the scanned images were analyzed with ImageJ (version 1.48), which is publicly available (<https://imagej.nih.gov/ij/download.html>, accessed 9 May 2018). The images were stacked by importing the image sequence.

A central area of 4900 mm² (70 by 70 mm²) in the cores was demarcated as the region of interest. The image area outside the region of interest was not considered for image analyses to exclude artificial voids close to the column walls and to minimize the impact of beam-hardening artifacts. We did not use any digital filter during image processing. Considering the limitations of CT resolution, we classified the imaged pores as macropores (Scott, 2000).

Soil structure in the sampled cores was formed by soil particles, loess dolls, living roots and decayed roots. Previously, Lamandé et al. (2013) and Duchesne et al. (2009) converted CT scanning images to extract a relative density. On the basis of the density difference of different parts in soil cores, we converted the original images into 8-bit grayscale images to distinguish air-filled pores from solids. We measured the grayscale values for each slice and calculated the mean grayscale value for each soil depth in the cores. With attenuation, the partial volume effect, and beam hardening, completely and perfectly discriminating roots is difficult. We considered soil particles, loess

dolls, and living roots as the soil matrix. The intensity value from the water phantoms (38–42, mean = 40) was used as the threshold value to differentiate air-filled spaces from the other regions. Values from 0 to 40 indicated air-filled pores and values from 40 to 255 indicated nonpores (Fig. 2). According to the intensity value of the soil pores, the root volume was not classified as a pore. CT-measured pores include pores formed by decayed roots and interaggregate macropores formed through repeated freezing and thawing or wetting and drying (Luo et al., 2010). After obtaining the segment pores from the soil matrix, the macropore parameters (macropore number, macropore volume fraction, branch density, connectivity density, and junction density) were obtained with the ImageJ software (Doubé et al., 2010; Li et al., 2016).

Physical and Chemical Properties of the Soil

Each soil core was cut into six zones after scanning (0–60, 60–120, 120–180, 180–240, 240–300, and 300–360 mm) to measure the physical and chemical properties of the soil. Soil samplers (5.3 cm long and 7 cm in diameter) were used to collect undisturbed samples from each depth. We then used a constant head method to determine K_{sat} . Sample cores were dried at 105°C for 24 h and weighed to determine the dry bulk density (BD) (g cm⁻³). The soil organic C (SOC, g kg⁻¹) and particle size distribution for each depth were also measured in the laboratory (Table 1). The disturbed soil samples were air-dried and divided into two subsamples. One subsample was ground to pass through a 1-mm mesh and was subjected to particle size analysis by laser diffraction with a

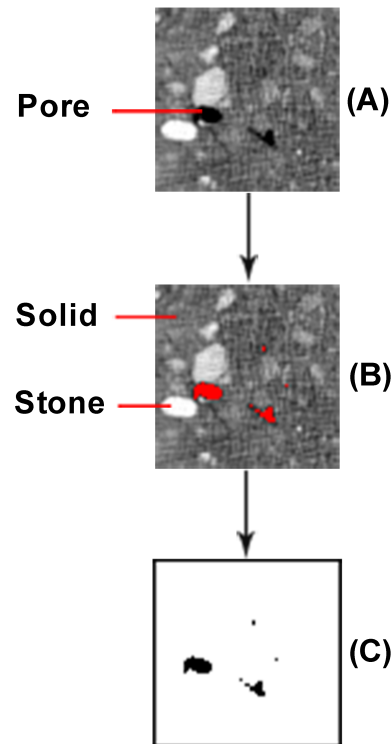


Fig. 2. Typical cross-sectional scanning image (A) and the segmentation of soil pores [(B) and (C)] used for data extraction. In (A), macropores are shown in black, solid particles are shown in gray, and stones are shown in white. After thresholding, the areas of air-filled pores are shown in red (B) and isolated in black (C).

Table 1. Mean ($\pm$ SD) soil organic C content (SOC), bulk density (BD), and sand and clay contents for the 0- to 60-, 60- to 120-, 120- to 180-, 180- to 240-, 240- to 300-, and 300- to 360-mm soil depths in the *Q. liaotungensis* (QLI), *P. tabuliformis* (PTA), *C. korshinskii* (CKO), and *M. sativa* (MSA) treatments.

Depth mm	Sand				Clay				BD				SOC			
	QLI	PTA	CKO	MSA	QLI	PTA	CKO	MSA	QLI	PTA	CKO	MSA	QLI	PTA	CKO	MSA
	%				%				g cm ⁻³				g kg ⁻¹			
0–60	18.7 $\pm$ 0.38	19.7 $\pm$ 0.48	51.7 $\pm$ 0.81	39.5 $\pm$ 0.09	13.2 $\pm$ 0.05	12.6 $\pm$ 0.04	9.4 $\pm$ 0.15	9.2 $\pm$ 0.02	0.88 $\pm$ 0.23	0.92 $\pm$ 0.08	1.26 $\pm$ 0.11	1.49 $\pm$ 0.28	34.3 $\pm$ 0.41	25.3 $\pm$ 0.67	8.3 $\pm$ 0.20	13.9 $\pm$ 0.42
60–120	16.7 $\pm$ 0.23	19.3 $\pm$ 0.32	47.9 $\pm$ 0.75	39.2 $\pm$ 0.30	13.1 $\pm$ 0.03	13.0 $\pm$ 0.03	9.5 $\pm$ 0.13	9.3 $\pm$ 0.05	0.91 $\pm$ 0.11	0.95 $\pm$ 0.28	1.30 $\pm$ 0.07	1.49 $\pm$ 0.29	24.1 $\pm$ 0.35	20.1 $\pm$ 0.36	5.0 $\pm$ 0.11	10.2 $\pm$ 0.56
120–180	14.0 $\pm$ 0.08	19.1 $\pm$ 0.16	45.2 $\pm$ 0.69	38.5 $\pm$ 0.52	14.0 $\pm$ 0.00	13.1 $\pm$ 0.01	9.6 $\pm$ 0.12	9.5 $\pm$ 0.08	1.05 $\pm$ 0.30	1.14 $\pm$ 0.10	1.29 $\pm$ 0.29	1.48 $\pm$ 0.22	20.8 $\pm$ 0.299	17.9 $\pm$ 0.06	3.8 $\pm$ 0.16	8.2 $\pm$ 0.69
180–240	12.4 $\pm$ 0.14	19.0 $\pm$ 0.12	39.9 $\pm$ 0.39	38.0 $\pm$ 0.53	14.6 $\pm$ 0.02	12.6 $\pm$ 0.01	9.9 $\pm$ 0.06	9.7 $\pm$ 0.11	1.05 $\pm$ 0.03	1.09 $\pm$ 0.24	1.28 $\pm$ 0.07	1.45 $\pm$ 0.29	15.2 $\pm$ 0.18	14.7 $\pm$ 0.04	3.2 $\pm$ 0.22	4.8 $\pm$ 0.41
240–300	10.6 $\pm$ 0.20	18.2 $\pm$ 0.09	37.3 $\pm$ 0.08	45.3 $\pm$ 0.55	15.4 $\pm$ 0.05	12.1 $\pm$ 0.01	10.6 $\pm$ 0.00	9.0 $\pm$ 0.13	1.11 $\pm$ 0.04	1.12 $\pm$ 0.21	1.31 $\pm$ 0.18	1.43 $\pm$ 0.10	10.9 $\pm$ 0.54	9.7 $\pm$ 0.02	3.2 $\pm$ 0.28	2.9 $\pm$ 0.12
300–360	10.6 $\pm$ 0.00	20.8 $\pm$ 0.15	34.7 $\pm$ 0.16	35.0 $\pm$ 0.98	14.8 $\pm$ 0.04	10.2 $\pm$ 0.02	13.0 $\pm$ 0.06	9.9 $\pm$ 0.17	1.20 $\pm$ 0.18	1.20 $\pm$ 0.01	1.32 $\pm$ 0.22	1.45 $\pm$ 0.07	9.4 $\pm$ 0.41	4.6 $\pm$ 0.02	3.3 $\pm$ 0.39	2.5 $\pm$ 0.01
Mean	13.8	19.4	42.8	39.2	14.2	12.3	10.3	9.4	1.03	1.07	1.29	1.47	19.1	15.4	4.5	7.1

Mastersizer 2000 (Malvern Instruments, Malvern, UK). The other subsample was passed through a 0.25-mm mesh for an analysis of SOC via the potassium dichromate–wet combustion method (Nelson and Sommers, 1982).

Data Analysis

We used the mean values with SD to show the soil parameters under different vegetational restoration. One-way ANOVA was conducted to compare the soil parameters among different treatments. Different regression analysis using SPSS version 16.0 (IBM Corp., Armonk, NY) were used to determine the relationships between K_{sat} and soil parameters. The regression model with the highest R^2 values was chosen to show the relationship between K_{sat} and soil parameters. Figures were computed with Origin version 8.0 software (OriginLab, Northampton, MA) and Photoshop CS version 6.0 (Adobe Systems Corporation, San Jose, CA).

RESULTS AND DISCUSSIONS

Computed Tomography-Measured Soil Properties under Different Vegetational Restoration Programs

Computed tomography-measured soil properties and the 3D visualization of macropores of the four treatments are listed

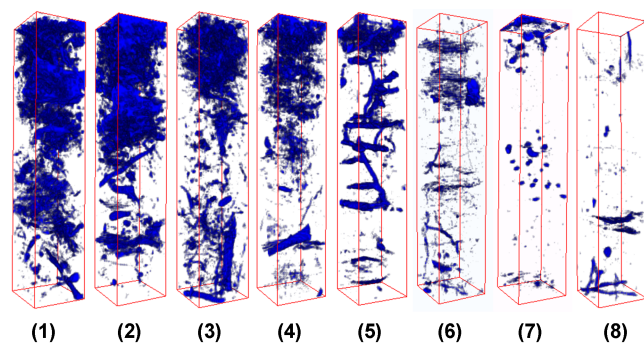


Fig. 3. Three-dimensional images of the 0-to 360-mm soil profile in the *Q. liaotungensis* [QLI: (1) and (2)], *P. tabuliformis* [PTA: (3) and (4)], *C. korshinskii* [CKO: (5) and (6)], and *M. sativa* [MSA: (7) and (8)] treatments. Macropores are shown in white and solid particles are shown in blue in the plotted plates.

in Fig. 3 and Fig. 4. Soil macropore volume fraction improved with the succession of vegetation. The two sample cores from MSA had the lowest macropore volume fraction and fewer continuous macropores than the other three treatments. The 3D images for QLI and PTA identified highly extensive and continuous macropores (Fig. 3). The mean volume fractions across the soil cores for the QLI, PTA, CKO, and MSA treatments were 6.6, 3.5, 1.3, and 0.6%, respectively (Fig. 4). In fact, the macropore volume fractions was equal to the macroporosity in this study. It is more variable than that in the soils in grass and tree buffers (1.5–3.5%) reported by Udawatta et al. (2006). The macropore volume fraction was found to be highest in QLI at each soil depth, except at the sixth depth. The macropore volume fraction was higher in PTA than in CKO but higher in CKO than in MSA at each depth (Fig. 4). Seobi et al. (2005) also found threefold higher total porosities to a depth of 76 mm in grass and agroforestry buffers than in a crop buffer. Zhao et al. (2010) found that soil associated with *Artemisia sacrorum* Ledeb., *Hippophae rhamnoides* L., and *Q. liaotungensis* has 0.9-, 1.7-, and 3.9-fold higher macroporosities in the 15- to 57-mm soil layer than soil in abandoned farmland. The macropore volume fraction also depended on soil depth. In our study, macroporosity was lowest in MSA, with values of 0 at the second, third, fourth, and fifth depths (Fig. 4). Macropore volume fraction in QLI decreased with depth, peaking at the first depth (0–60 mm) and decreasing by 85% at the sixth depth (300–360 mm). By contrast, the variation of the macropore volume fraction with depth in PTA, CKO, and MSA was irregular and was not the lowest at the sixth depth (Fig. 4).

The QLI, PTA, and CKO cores had means of 52, 36, and 16 macropores, respectively, and MSA had only six macropores within 4900 mm². The number of macropores for the grass samples (six) in our study was lower than the number (14) found by Udawatta et al. (2006) and was also lower than that (12–78) in an Inner Mongolia grassland (Hu et al., 2015). Macropore numbers in each depth were all higher for QLI and PTA than for CKO, followed by MSA (Fig. 4). The QLI treatment had 100,

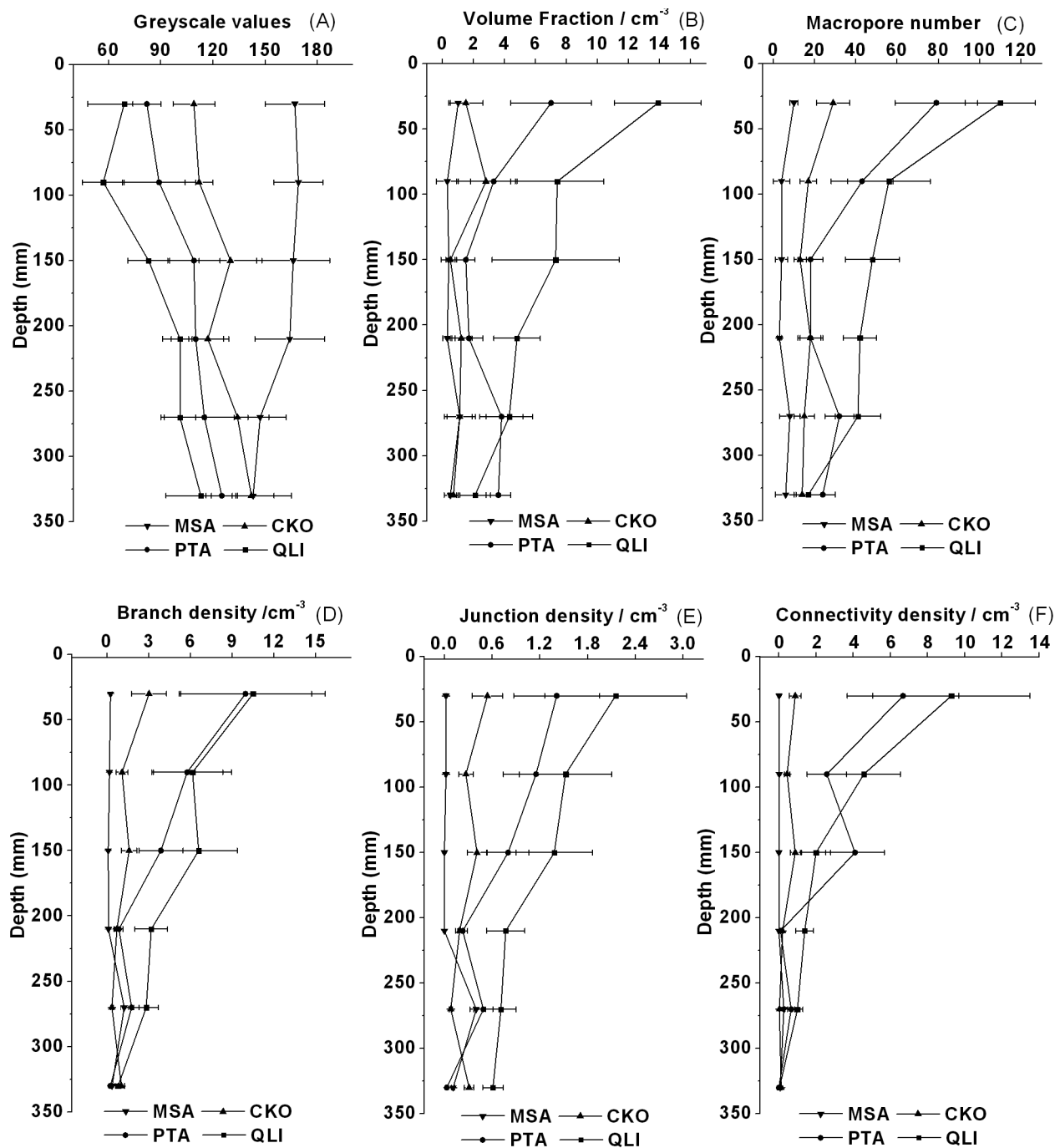


Fig. 4. Greyscale values (A), macropore volume fraction (B), macropore number (C), branch density (D), junction density (E), and connectivity density (F) derived from computed tomography (CT) images for the 0- to 60-, 60- to 120-, 120- to 180-, 180- to 240-, 240- to 300-, and 300- to 360-mm soil depths and mean values with SD under the *Q. liaotungensis* (QLI), *P. tabuliformis* (PTA), *C. korshinskii* (CKO), and *M. sativa* (MSA) treatments in the Loess Plateau of China.

52, 44, 38, 34, and 11 more macropores than MSA at the sixth depth from the shallowest to the deepest (Fig. 4), with the difference decreasing with depth. Macropore number was highest at the first (0–60 mm) depth for all treatments; the order was QLI > PTA > CKO > MSA (Fig. 4).

The high macropore branch and junction densities suggested an extensive, well-connected, and complex pore network. Luo et al. (2010) argued that the characteristics of macropore networks depend on soil type and land use; for example, pasture soils have higher macroporosity, length density, and node density

than soils in a row crop. In our study, the 3D macropore parameters depended on vegetation type and soil depth. The connectivity density, branch density, and junction density of the macropores were in the order QLI > PTA > CKO > MSA. The values in QLI and PTA were higher at the first depth (0–60 mm) than at greater depths, and the values in QLI decreased with depth. The values in CKO and MSA, however, were not the highest in the first depth (Fig. 4). Connectivity and junction density at the third and fourth depths were notably near 0, indicating that macropore connectivity was lowest at these depths.

We measured the grayscale value of each slice and calculated the mean grayscale value for each soil depth in different cores. On average, the grayscale value was highest (159) in MSA, followed by CKO (124), PTA (105), and QLI (87). The grayscale value was lower in CKO than in MSA but was higher in CKO than in QLI and PTA throughout the soil profile. The differences between QLI and PTA, however, were not large, especially at the two deeper depths (240–360 mm) (Fig. 4). The grayscale value also slightly increased with depth in QLI, PTA, and CKO but not in MSA. A significantly ($P < 0.001$) regressive relationship was observed between the grayscale value and BD with a high R^2 of 0.9032 (Fig. 5). We considered the grayscale values to be feasible to use to distinguish the difference in BD. Plant roots can reduce soil BD and improve soil structure through root penetration (Mishra et al., 2003; Udawatta et al., 2006; Jiang et al., 2007) and thus contribute to the differences in the CT-measured grayscale values of soil cores under different cover vegetation.

Saturated Hydraulic Conductivity under Four Perennial Vegetation Types

Figure 6 shows the values of K_{sat} for the treatments and depths. On average, K_{sat} was the highest in QLI (0.073 cm min^{-1}), followed by PTA (0.063 cm min^{-1}) and CKO (0.056 cm min^{-1}), and was lowest in MSA (0.018 cm min^{-1}). Similar results were reported by the study of Liang et al. (2009), who observed that the average K_{sat} in the 60-cm soil profile significantly increased from bare land to shrubs to forest. *Quercus liaotungensis* supported the highest values under the nine stands, with vegetative restoration notably improving K_{sat} . Moreover, in the shrub stage, the average K_{sat} showed a significant difference from that of the bare land. In this study, the values of K_{sat} were much higher under QLI, PTA, and CKO than under MSA at all depths. When the succession went into the climax stage, the average K_{sat} reached the maximum value (0.328 cm min^{-1}) (Peng et al., 2010) which was much higher than that of QLI (0.073 cm min^{-1}) in our study. This result may be caused by the significant differences in the soil properties (soil clay contents, BD, and porosity) in these two studies. In addition, K_{sat} slightly decreased with depth un-

der all treatments (Fig. 6). The average values of K_{sat} at the first depth under QLI, PTA, CKO, and MSA treatments were 0.090, 0.074, 0.065, and 0.027, respectively, which are all higher than those at the sixth depth (0.054, 0.053, 0.045, and 0.020, respectively). This result is consistent with those of previous studies by Liang et al. (2009) and Peng et al. (2010). All the observations suggest that the vegetation succession process can significantly affect K_{sat} .

Predicting K_{sat}

Many studies have reported a positive correlation between K_{sat} and CT-measured macropore parameters (Udawatta et al., 2006, 2008; Kim et al., 2010; Kumar et al., 2010; Anderson, 2014). Macropores constituting only 1.26 to 8.55% of the total macroporosity strongly accounted for the variability in K_{sat} (Shipitalo et al., 2000; Hayashi et al., 2006). Lu et al. (2014) studied the distribution of soil macropores in the Huoditang Forest region and argued that the density of macropores accounted for 84% of the variability in K_{sat} . Poor soil structure probably contributed to low rates of water transmission (Pachepsky et al., 2000). In our study, single macropore parameters (volume fraction, number, branch density, connectivity density, and junction density) could explain 50.2 to 71.3% of the variation in K_{sat} . The conduction of water through macropores strongly depends on the 3D geometry and topology of the macropores. Macropore continuity significantly increases water flow and solute transport in soil (Luo et al., 2008). However, in this study, macropore quantity (volume fraction and number) had a higher R^2 for predicting K_{sat} than the morphology of the macropores (branch density, connectivity density, and junction density) (Table 2). Moreover, the grayscale values were negatively correlated with K_{sat} and accounted for 78.8% of the variation in K_{sat} . Grayscale values plus macropore number accounted for 81.2% of the variation in K_{sat} .

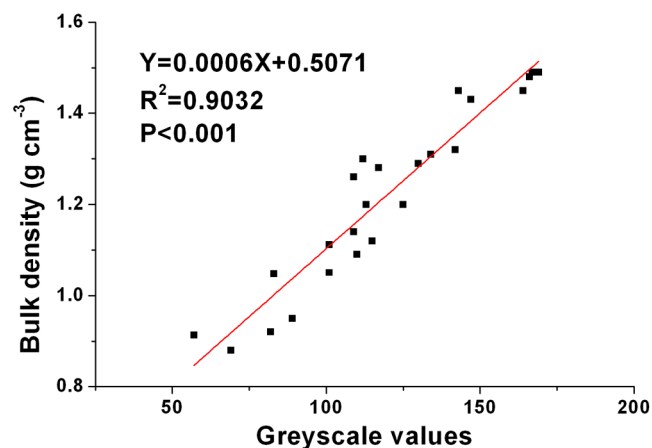


Fig. 5. Relationship between saturated hydraulic conductivity and computed tomography (CT)-measured grayscale values.

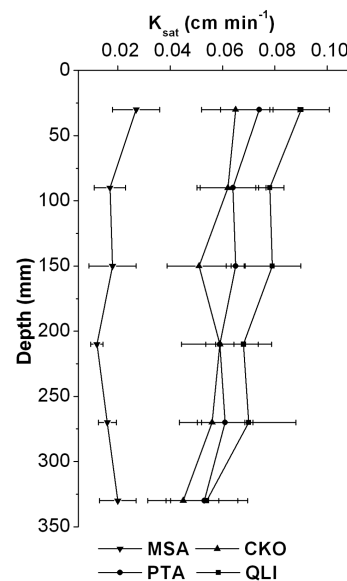


Fig. 6. Mean saturated hydraulic conductivity (K_{sat}) for the 0- to 60-, 60- to 120-, 120- to 180-, 180- to 240-, 240- to 300-, and 300- to 360-mm soil depths under the *Q. liaotungensis* (QLI), *P. tabuliformis* (PTA), *C. korshinskii* (CKO), and *M. sativa* (MSA) treatments.

Table 2. Relationships between saturated hydraulic conductivity (K_{sat}) and computed tomography (CT)-measured parameters, soil texture (clay, silt, and sand), soil organic C content (SOC), and bulk density (BD).

Parameters	N	R^2	P-value
Single parameters			
$K_{\text{sat}} = -0.0007(\text{grayscale values}) + 0.1325$	48	0.788	0.001
$K_{\text{sat}} = 0.0404 + 0.0195 \times \ln(\text{volume fraction})$	48	0.713	0.001
$K_{\text{sat}} = -0.0135 + 0.0225 \times \ln(\text{macropore number})$	48	0.692	0.001
$K_{\text{sat}} = 0.05 + 0.013 \times \ln(\text{branch density})$	48	0.601	0.001
$K_{\text{sat}} = 0.068 + 0.011 \times \ln(\text{junction density})$	48	0.555	0.001
$K_{\text{sat}} = 0.062 + 0.007 \times \ln(\text{connectivity density})$	48	0.502	0.001
$K_{\text{sat}} = 0.1778 - 0.1038(\text{BD})$	48	0.769	0.001
$K_{\text{sat}} = 0.05 + 0.015 \times \ln(\text{SOC})$	48	0.546	0.001
$K_{\text{sat}} = 0.05 + 0.015 \times \ln(\text{silt})$	48	0.546	0.001
$K_{\text{sat}} = 0.064 + 0.008 \times \ln(\text{Clay})$	48	0.473	0.001
$K_{\text{sat}} = 0.068 + 0.012 \times \ln(\text{Sand})$	48	0.450	0.001
Multiple parameters			
$K_{\text{sat}} = 0.027011 - 0.00029 \times (\text{grayscale values}) - 0.00795 \times \ln(\text{volume fraction}) + 0.022156 \times \ln(\text{macropore number})$	48	0.819	0.001
$K_{\text{sat}} = 0.042898 - 0.00027 \times (\text{grayscale values}) + 0.01425 \times \ln(\text{macropore number})$	48	0.812	0.001
$K_{\text{sat}} = -0.02941 - 0.00634 \times \ln(\text{volume fraction}) + 0.029282 \times \ln(\text{macropore number})$	48	0.797	0.001
$K_{\text{sat}} = 0.102402 - 0.00046 \times (\text{grayscale values}) + 0.007076 \times \ln(\text{volume fraction})$	48	0.779	0.001

Grayscale values, volume fraction, and macropore number were the best combination of CT-measured parameters for predicting K_{sat} , accounting for 81.9% of the variation in K_{sat} (Table 2). Udawatta and Anderson (2008) found a significant linear regression relationship between pore number and $\log(K_{\text{sat}})$. Kim et al. (2010) argued that a linear regression model could be used to estimate K_{sat} with macroporosity. Anderson et al. (2010) found a significant linear regression relationship between $\log(K_{\text{sat}})$ and $\log(\text{pore number})$. In the present study, we tried different regression models to predict K_{sat} and chose the model with a high R^2 to show the regression relationship between soil properties and K_{sat} (Table 2). Grayscale values and BD exhibited a linear regression relationship with K_{sat} . For the other parameters, we used a logarithmic model to predict K_{sat} . A linear function combined with logarithmic function revealed a high R^2 in the two- or three-parameter regression equations (Table 2).

Bulk density is the key factor affecting the spatial distribution of K_{sat} in sloping land (Zhao et al., 2013). In the present study, BD was negatively correlated with K_{sat} and accounted for most of the variation (76.9%) as a single variable. Bulk density, as a single variable, accounted for only 22% of its variability in the study by Udawatta et al. (2008). High levels of SOC can contribute to improved soil porosity and high K_{sat} (Udawatta and Anderson, 2008; Peng et al., 2010). As a single variable, SOC explained 64.6% of the variation in K_{sat} . Soil silt content explained 54.6% of the variation in K_{sat} , followed by soil clay content (47.3%) and sand content (45.0%). Traditional methods have been used in many studies to evaluate soil structure and K_{sat} in the Loess Plateau (Shi et al., 2007; Yu et al., 2013; Zhao et al., 2013). Although traditional methods can be used to evaluate the influences of management on changes in soil properties, CT scanning is better than traditional methods when measuring soil properties in great detail and estimating K_{sat} .

CONCLUSION

The values of CT-measured macropore parameters for the four treatments were in the order $\text{QLI} > \text{PTA} > \text{CKO} > \text{MSA}$. Vegetation recovery was shown to increase K_{sat} by improving soil architecture. Grayscale values, macropore volume fraction, and macropore number best predicted K_{sat} as single parameters, and they collectively accounted for 81.9% of the variation in K_{sat} . Macropore quantity (volume fraction and macropore number) showed a higher R^2 for predicting K_{sat} than macropore morphology (branch density, connectivity density, and junction density). Computed tomography scanning can provide higher accuracy for estimating K_{sat} than traditional methods. The results of the study show that the characteristics of soil macropores in the upper root zone varied significantly with the recovery of different perennials. The CT-measured soil macropore parameters could be used to estimate K_{sat} .

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